

PGR106

Linear Algebra

Lecture 9 – Singular Value Decomposition
& Least Squares Problem



Singular Value Decomposition

- **Singular Value Decomposition** (**SVD**) is a way of factorizing a matrix into singular vector and singular values.
- **SVD** is one of the most important concepts of data science and is used in many applications such as **dimensionality reduction**, **noise reduction**, and **data analysis**.
- **SVD** can be computed for any real matrix (square or rectangular).
- **SVD** is the product of three matrices.

$$A = U \Sigma V^T$$

$$A \rightarrow m \times n \quad U \rightarrow m \times m \quad \Sigma \rightarrow m \times n \quad V \rightarrow n \times n$$

Singular Value Decomposition

$$A = U \Sigma V^T$$

$$A \rightarrow m \times n \quad U \rightarrow m \times m \quad \Sigma \rightarrow m \times n \quad V \rightarrow n \times n$$

- U and V are both orthogonal matrices.
- Columns of matrices U and V are unit vectors and orthogonal to each other.

Singular Value Decomposition

$$A = U \Sigma V^T$$

$$A \rightarrow m \times n \quad U \rightarrow m \times m \quad \Sigma \rightarrow m \times n \quad V \rightarrow n \times n$$

Left Singular Vectors $U = [u_1 \quad u_2 \quad \cdot \quad \cdot \quad u_m]$

Right Singular Vectors $V = [v_1 \quad v_2 \quad \cdot \quad \cdot \quad v_n]$

$$\Sigma = \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_m \end{bmatrix}$$

Singular Values

Singular Value Decomposition

$$A = U \Sigma V^T$$

Left Singular Vectors

$$U = [u_1 \quad u_2 \quad \cdot \quad \cdot \quad u_m]$$

Eigenvectors of AA^T

Right Singular Vectors

$$V = [v_1 \quad v_2 \quad \cdot \quad \cdot \quad v_n]$$

Eigenvectors of $A^T A$

$$\Sigma = \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_m \end{bmatrix}$$

Singular Values

$$\sigma_i = \sqrt{\text{eigenvalues of } A^T A}$$

$$\sigma_1 > \sigma_2 > \cdots > \sigma_m$$

Singular Value Decomposition

- Steps for computing **SVD**

$$A = U \Sigma V^T$$

- ✓ Compute the **eigenvalues** and **eigenvectors** of AA^T to get U
- ✓ Compute the **eigenvalues** and **eigenvectors** of $A^T A$ to get V
- ✓ Compute the singular values as the square roots of the eigenvalues of $A^T A$

Singular Value Decomposition

Exercise 1

Find the SVD of A both by hand and in Python.

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

Singular Value Decomposition

Exercise 2

Find the SVD of A in Python.

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Least Squares Problem

- Linear system is a collection of one or more linear equations involving the same variables.

$$a_1x_1 + a_2x_2 + \cdots \dots + a_nx_n = b$$

Linear Systems of n Unknowns

$$x_1 \rightarrow ? \quad x_2 \rightarrow ? \quad x_3 \rightarrow ? \quad x_n \rightarrow ?$$

Least Squares Problem

$$\begin{aligned}
 a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\
 a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\
 \dots &= \dots \\
 a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m
 \end{aligned}$$

$$Ax = b$$

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$m \times n \qquad n \times 1 \qquad m \times 1$

Least Squares Problem

$$\begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & \cdot & \cdot & a_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{m1} & a_{m2} & \cdot & \cdot & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \cdot \\ \cdot \\ b_m \end{bmatrix}$$

$m \times n \qquad n \times 1 \qquad m \times 1$

$$Ax = b$$

- **Case 1:** number of equations is equal to the number of variables ($m = n$)
- **Case 2:** number of equations is less than the number of variables ($m < n$)
- **Case 3:** number of equations is more than the number of variables ($m > n$)

Least Squares Problem

Inconsistent $\begin{cases} x_1 = 1 & \longrightarrow \textcircled{1} \\ 2x_1 = -\frac{1}{2} & \longrightarrow \textcircled{2} \\ 3x_1 + x_2 = 5 & \longrightarrow \textcircled{3} \end{cases}$

No Solution

$$Ax = b$$

$$\begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1/2 \\ 5 \end{bmatrix}$$

$m \times n$ $n \times 1$ $m \times 1$

Equations are more than number of variables: $m > n$

Least Squares Problem

$$Ax = b$$

$$Ax - b = 0$$

$$Ax - b = r$$

close to 0

Residual

$$Ax \approx b$$

Least Squares Problem

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$Ax = b$$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$



$$Ax - b = 0$$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$Ax - b = r$$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix}$$

Least Squares Problem

$$Ax - b = r$$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix}$$

Minimize

$$\text{Minimize: } r_1^2 + r_2^2 + r_3^2 \rightarrow 0$$

$$\begin{bmatrix} r_1 & r_2 & r_3 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix}$$

$$\text{Minimize: } (Ax - b)^T (Ax - b)$$

Least Squares Problem

$$\min [(Ax - b)^T (Ax - b)]$$

$$(Ax)^T - b^T$$

$$(x^T A^T - b^T) (Ax - b)$$

$$x^T A^T Ax - x^T A^T b - b^T Ax + b^T b$$

$$(Ax)^T b$$

$$x^T A^T Ax - b^T Ax - b^T Ax + b^T b$$

$$x^T A^T Ax - 2b^T Ax + b^T b$$

Transpose Properties:

$$(A - B)^T = A^T - B^T$$

$$(AB)^T = B^T A^T$$

$$A^T B = B^T A$$

Least Squares Problem

$$\min [(Ax - b)^T (Ax - b)]$$

$$x^T A^T A x - 2b^T A x + b^T b$$



$$(Ax)^T Ax$$



$$A^T A x x$$



$$A^T A x^2 - 2b^T A x + b^T b$$

Least Squares Problem

Optimization:

$$\min [(Ax - b)^T (Ax - b)]$$

$$\min [\underbrace{A^T A x^2 - 2b^T A x + b^T b}_{f(x)}]$$

$$f(x)$$

Solution is obtained by differentiating $f(x)$ with respect to x and differential to 0

$$\frac{\partial f(x)}{\partial x} = 0$$

Least Squares Problem

Optimization:

$$\min [(Ax - b)^T (Ax - b)]$$

$$\min [\underbrace{A^T A x^2 - 2b^T A x + b^T b}_{f(x)}]$$

$$f(x)$$

$$\frac{\partial f(x)}{\partial x} = 0 \quad \frac{\partial (A^T A x^2 - 2b^T A x + b^T b)}{\partial x} = 0$$

$$2A^T A x - 2b^T A = 0$$

$$2A^T A x - 2A^T b = 0$$

$$A^T A x = A^T b$$

$$x = (A^T A)^{-1} A^T b$$

Least Squares Problem

For example:

$$x_1 = 1$$

$$2x_1 = -\frac{1}{2}$$

$$3x_1 + x_2 = 5$$

$$Ax = b$$

$$\begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -0.5 \\ 5 \end{bmatrix}$$

Optimization:

$$x = (A^T A)^{-1} A^T b$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \left[\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \right]^{-1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -0.5 \\ 5 \end{bmatrix}$$

Least Squares Problem

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \left[\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \right]^{-1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -0.5 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 14 & 3 \\ 3 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 15 \\ 5 \end{bmatrix}$$

Solution:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.2 & -0.6 \\ -0.6 & 2.8 \end{bmatrix} \begin{bmatrix} 15 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

Least Squares Problem

$$\begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -0.5 \\ 5 \end{bmatrix}$$

Optimization:

$$x = (A^T A)^{-1} A^T b$$

Solution:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

Best Solution possible

$$\begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} \neq \begin{bmatrix} 1 \\ -0.5 \\ 5 \end{bmatrix}$$

End of Lecture 9



MATHEMATICS FOR MACHINE LEARNING

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